Topic

Bachelor of Technology
Computer Science and Engineering

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JANUARY 2026

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1. Introduction

Research is a systematic way of finding answers to questions. It involves collecting information, analyzing it, and reaching a conclusion. We categorize research based on *why* we are doing it (purpose), *how* we are doing it (method), and *what kind of data* we use.

2. Classification by Purpose (Why are we doing it?)

This category looks at the goal of the study.

- **Basic (Fundamental) Research:**
    - **Goal:** To gain knowledge for the sake of knowledge. It tries to understand how things work without worrying about immediate practical use.
    - **Example:** A study on "How the human brain stores memories."
- **Applied Research:**
    - **Goal:** To solve a specific, practical problem facing a society or business.
    - **Example:** A study to find "The most effective cure for a specific disease."

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3. Classification by Data Type (What info are we using?)

This category looks at the type of data being collected.

- **Quantitative Research (Numbers):**
    - **Focus:** Deals with numbers, statistics, and things you can measure.
    - **Methods:** Surveys, polls, and experiments.
    - **Example:** "What percentage of students prefer online classes over offline classes?"
- **Qualitative Research (Words):**
    - **Focus:** Deals with descriptions, feelings, opinions, and meanings. It helps understand *why* something happens.
    - **Methods:** Interviews, focus groups, and observations.
    - **Example:** "How do students *feel* about the stress of online exams?"

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4. Classification by Depth (How much do we know?)

This looks at how much information we already have on the topic.

- **Exploratory Research:**
    - **Goal:** Done when a problem is not clearly defined. It helps you explore a new topic.
    - **Example:** A company looking into a brand new market they know nothing about.
- **Descriptive Research:**
    - **Goal:** describes the characteristics of a population or phenomenon. It answers "What," "Where," "When," and "How," but not "Why."
    - **Example:** A census report describing the age and gender of people in a city.
- **Explanatory (Causal) Research:**
    - **Goal:** Explains the cause-and-effect relationship between variables. It answers "Why."
    - **Example:** "Why does smoking cause lung cancer?"

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5. Classification by Method

- **Experimental Research:**
    - **Goal:** The researcher changes one variable (independent) to see the effect on another variable (dependent) while keeping everything else the same.
    - **Example:** Testing if a new fertilizer makes plants grow taller than the old fertilizer.
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- **Correlational Research:**
    - **Goal:** Looks for a relationship between two things without the researcher changing anything.
    - **Example:** Studying if there is a link between the amount of ice cream sold and the number of sunglasses sold. (They are related, but one doesn't cause the other!).

6.Conclusion

Choosing the right type of research depends on what you want to find out. Often, researchers use a mix of these methods (like Mixed Research) to get the best results.

7.References

1. **Kothari, C. R. (2004).** *Research Methodology: Methods and Techniques*. New Age International.
    - *Note: This is the standard book for engineering (B.Tech) students in India.*
2. **Creswell, J. W. (2014).** *Research Design*. SAGE Publications.